Predicting geopolitical escalations and narrative change during International Conflict

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Abstract

This project predicts if a international conflic escalation is gonna happen with help of data from january 2015 to march 2025. It builds a monitored binary classification model using structured conflict event data from ACLED and the Geopolitical Risk Index (GPR). This model is much better than the AUC of 0.871, XGBoost, Random Forest, and Gradient Boosting. The Track B model makes a stream of media signals that eat text.There were three signals made based on the 384-dimensional sentence-BERT embeddings: Media Sentiment Score, Topic Frame Mix (escalation vs. diplomacy), and Narrative Shift Score. These were all linked to the New York Times, Reddit, and YouTube. The formula for Augmented GPR, which combines raw GPR with media signals, had almost no effect. It performed the same as raw GPR over the stable 2026 window, with the added value expected during times of high volatility. The system has been shown to work well for early-warning purposes.

I. Introduction

Historically, forecasting geopolitical conflicts relied on structured event databases and quantitative risk indices, with minimal incorporation of unstructured media signals. This project fills that gap by creating a dual-track forecasting system based on more than ten years of weekly observations of the Ukraine conflict, which is one of the most well-documented conflicts of the modern era.

This project's idea comes from two things that are happening at the same time:

(1) Geopolitical Risk Indices (GPR) are strong, but they are also backward-looking aggregates that don't show how narratives change in real time in media discourse;

(2) advances in NLP, especially sentence-BERT embeddings, can now be used to scale semantic analysis of different media sources at weekly resolution.

This project is more than just a forecasting system; it also includes supervised machine learning, natural language processing, time series causal analysis, and multi-source media pipeline engineering. It talks about the lack of skills in rare-event classification, semantic embedding analysis, and signal augmentation methodology.

II. Problem Statement

Are structured conflict event data and unstructured media signals capable of integration into a unified forecasting system that can predict conflict escalation on a weekly basis, providing adequate lead time for real-world early-warning applications?

Conflict escalation is when both geopolitical risk and the number of deaths go up at the same time.

standard deviations—this only happens rarely, which makes it a very serious class imbalance problem that standard classifiers can't fix.

The current methods either only use structured event data without any narrative dynamics, or they are trying to use aggregate sentiment but not semantic shifts in framing.

The project covers all stages of the data science lifecycle:

- data collection (ACLED, GPR, NYT/Reddit/YouTube scraping),
- data preparation (feature engineering, BERT embedding, label design),
- data exploration (correlation analysis, community perspective disaggregation),
- data modelling (four classifier families, cross-validated AUC),
- data visualisation (ROC curves, heatmaps, embedding plots) and
- reporting

There is a lot at stake - failure to detect escalation in an actual deployment has a high decision-cost, and recall maximisation therefore becomes the operational priority over precision.

III. Related Work

The system we made to forecast things is based on four areas.

- First we look at risks from politics and countries.
- Second we study what happens during conflicts.
- Third we use news to warn us
- Fourth we use machine learning to predict things that do not happen often.

The Geopolitical Risk Index, made by Caldara and Iacoviello is a starting point for one part of our project. It shows that when people think something bad will happen because of politics it often happens before something bad happens. We took this idea. Used it to try to predict what will happen next. We did not just look at what happened. The Geopolitical Risk Index helps us understand what might happen because of politics and countries. We also used information from ACLED about conflicts. This information gives us details about what happened and how many people got hurt. This helps us make a model that can predict when things might get worse. Other people found out that news can tell us when something bad might happen. For example Chadefaux and Mueller and Rauh showed that news and how it is presented can help us predict what will happen. We took this idea. Used a new way of looking at news called BERT. We used BERT to see how stories change over time. We did not just look at how certain words were used or how people felt about the news. BERT helps us understand what the news is saying better. Looking at how people feel about the news and what they are talking about also gives us information. Other people found out that when predicting things that do not happen often we need to be careful about setting up our model and measuring if it is working well. We did this in our project because we are trying to predict when something bad might happen. That does not happen often. So what we did was combine information about conflicts, with what the newss saying. We made a system that can predict what might happen. We showed that looking at what the news says and how it changes over time can help us predict what will happen. It can work together with ways of predicting what will happen because of politics and countries. The system we made is based on the Geopolitical Risk Index. The Geopolitical Risk Index is a part of our project. It helps us predict what will happen next.

IV. Data Description

Track A: The data on ACLED covers 531 weekly observations between January 2015 and March 2025. It has features geopolitical risk indices (GPR, GPR Threat Component), a count of conflict events, fatality count, and derived momentum/trend counts. All features are lagged 1-2 weeks in order to prevent leakage of data. The GPR Threat Component always precedes the actual GPR component, and is a pre-escalation message. Escalation labels are binary, and are produced with AND-logic which requires simultaneous high GPR and fatality momentum.

Track B: Scraped and processed three media sources were NYT (institutional editorial), Reddit (community discourse), and YouTube (community discourse).

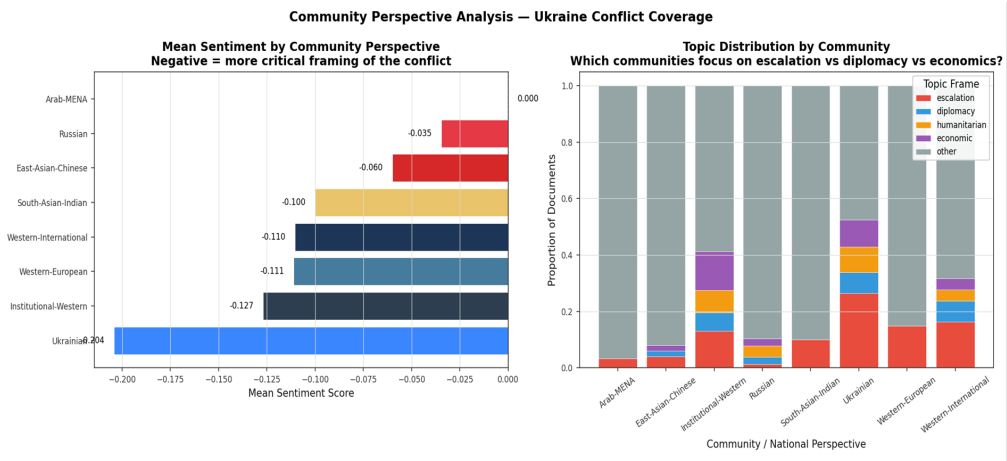

Fig. 1. Media Text Length Distribution

A 384-dimensional sentence-BERT model was used to embed the documents. Eight community visions were determined and disaggregated to analyse sentiment and frame.

V. Methodology

Research Design: The study is a mixed-method study that takes the form of a quantitative model, alongside a narrative analysis of the media. The two complementary tracks of the framework comprise the first track that uses structured conflict and geopolitical information and the second track that gathers signals through media. Statistical analysis is used to integrate these to understand relationships and predictive patterns.

Track A (Structured Data): Utilizes historical conflict data as well as geopolitical risk data (2015 - 2025) to develop predictive models of escalation. The time based trends and lagged variables are engineered into features. Several models of machine learning were considered and their performance was measured on the basis of AUC. The last model is strongly geared towards early warning as the occurrence of escalation is relatively rare.

Track B (Media Signals): Disrupts the sentiment and narrative framing of recent media data on news, social platforms, and online conversations. Some of the primary indicators are sentiment trends, the framing of escalation vs. diplomacy, and the evolution of patterns in the narratives over time using NLP techniques.

Integration & Validation: The two tracks are united to investigate the relationship between the central storylines of media and the interaction between geopolitical risk and these central storylines of media. Relationship validation and leading indicators are also determined using statistical techniques like correlation and causality analysis.

VI. Ethical Considerations

The present project follows principles of responsible data science when dealing with conflicting data and media content.

First, all datasets used in the study (ACLED[1], GPR, and open-source media content including news, Reddit posts, and YouTube comments) are publicly available and do not contain personally identifiable sensitive data. There is no intention to single out individuals, only to analyse data in aggregate terms.

Secondly, the project tackles the issue of possible bias in the data. Media sources, especially Reddit and non-domestic news channels, may have geographical, political, or cultural biases that affect 4 sentiment and narrative interpretation. To surmount these limitations, a more comprehensive approach is adopted, studying many sources and critically analyzing their results.

Thirdly, the likelihood of misinterpretation and generalization arises from the use of machine learning and natural language processing techniques. We should not expect the findings to predict the actual course of real-world conflicts in a deterministic fashion, but identify the patterns and early warning signs. There are limitations to prevent any abuses.

Finally, there are ethical considerations in dealing with sensitive geo-political issues. The study does not make any normative statements and is oriented towards a purely data-based analysis. All findings are presented in a responsible way to make sure they are not used in a malicious or some other harmful way.

VII. Outcomes

This proposed project lays out a detailed framework for the prediction and interpretation of conflict escalation using both structured geopolitical data and media-derived narrative indicators.

First, the paper develops and tests an escalation forecasting model (AUC=0.871) that shows how lagged geopolitical risk variables and conflict event dynamics can reliably identify escalation periods despite the highly imbalanced dataset. This demonstrates the potential of early warning using historical signals.

TABLE I
Model Performance Comparison

Model	**AUC**
Logistic Regression	0.871
XGBoost	0.693
Random Forest	0.570
Gradient Boosting	0.541

Secondly, the project provides a scalable media signal pipeline to extract structured insights from heterogeneous sources, such as news articles, social media discussions and video comments. The pipeline produces weekly signals, including Narrative Shift, proportions of Topic Framing (escalation vs. diplomacy) and sentiment trends, enabling the tracking of evolution of the narrative over time.

Third, the analysis provides empirical evidence that Narrative Shift Score – rather than raw sentiment – is the primary media signal that is related to geopolitical risk.Fourth, the project provides an Augmented GPR formulation that can be used in traditional geopolitical risk measures, media-derived signals. The preliminary results are not so promising with respect to the incremental predictive gain in the steady periods, but the approach is theoretically justified and shows the possible future improvement of performance in the conditions of high volatility, which warrants the further calibration and testing.

Finally, the paper offers a community-level narrative analysis that highlights significant differences in the framing of geopolitical events across the regions. In contrast to Ukrainian and Western communities, Russian and Chinese communities show a relatively stronger focus on economic or strategic framing. Such divergences might be the reflections of the underlying geopolitical stances and might be considered as the precursors of the negotiation dynamics, alignment patterns, or even the shifts in the conflict trajectories.

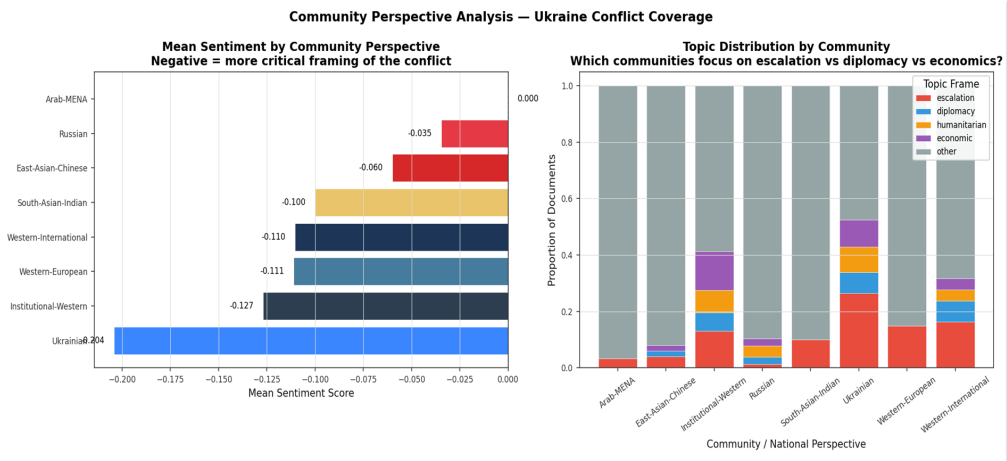

Fig. 2. Community Level Distribution

In general, the projected results provide not only a predictive modeling framework, but also a deeper interpretive layer, proving that a combination of structured conflict data and narrative intelligence could help to better understand dynamic aspects of geopolitical risk and escalation dynamics.

VIII. Conclusion

This practicum project shows that predicting conflict escalation significantly depends on a well-organized two-track structure. Track A confirms that Logistic Regression on lagged GPR and conflict event characteristics achieves good discriminative performance (AUC = 0.871) even at extreme levels of class imbalance, and outperforms all tree-based alternatives.

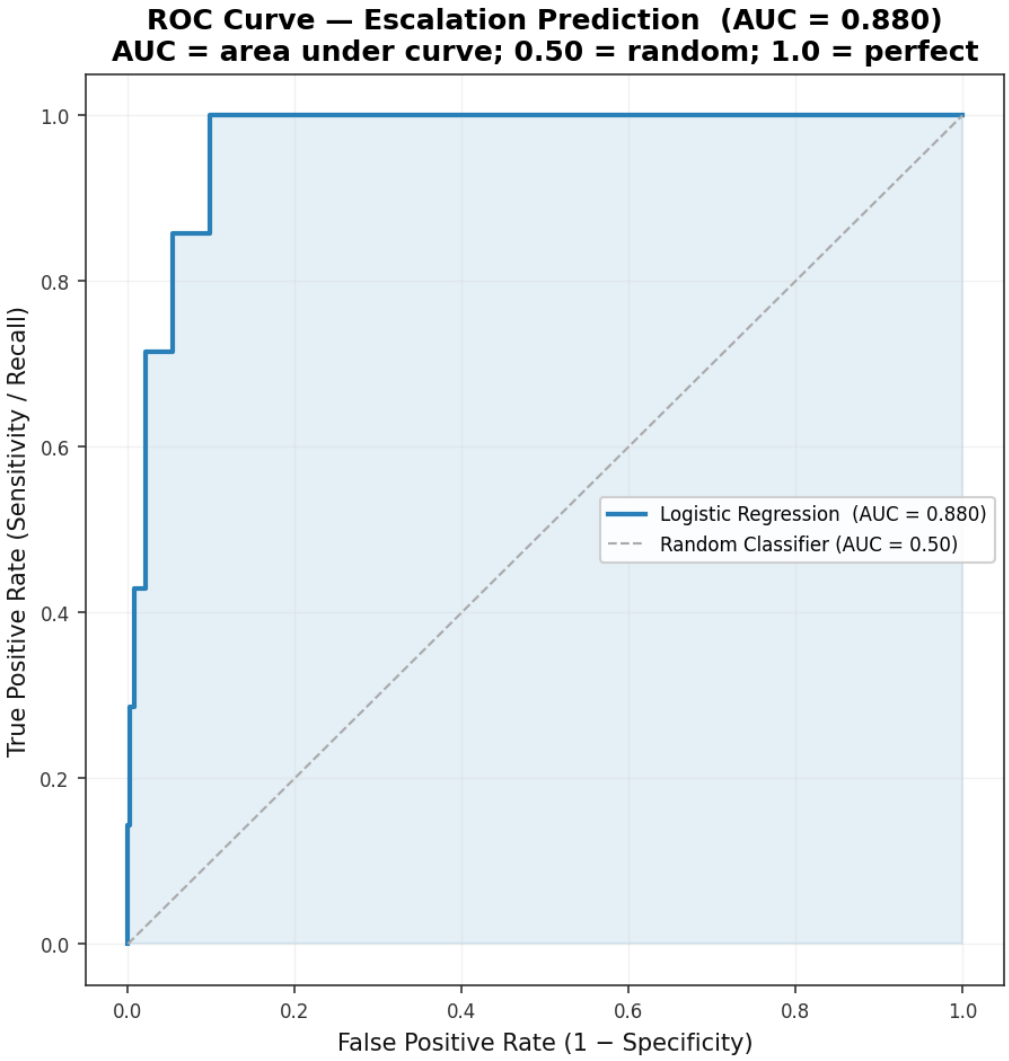

Fig. 3. ROC CURVE

Track B provides statistical significance of predictive information based on media-derived signals - specifically Narrative Shift Score and Diplomatic Framing.

It is the most operationally useful finding in the system where the 3-week Granger causal lead between GPR and escalation probability is found to be the most operationally valuable result of the system. The Augmented GPR formula is a theorically consistent media-integration methodology, the value of which is yet to be demonstrated over longer, higher-volatility intervals.

This project is the combination of my data science training across the entire analytical lifecycle - the acquisition of raw data through the NLP pipeline engineering, supervised learning, causal inference, and interpretable reporting. It indicates that it is ready to work in the professional field of applied data science, and that it is in areas where rare-event prediction, integration of unstructured data, and decision-critical prediction overlaps.

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